

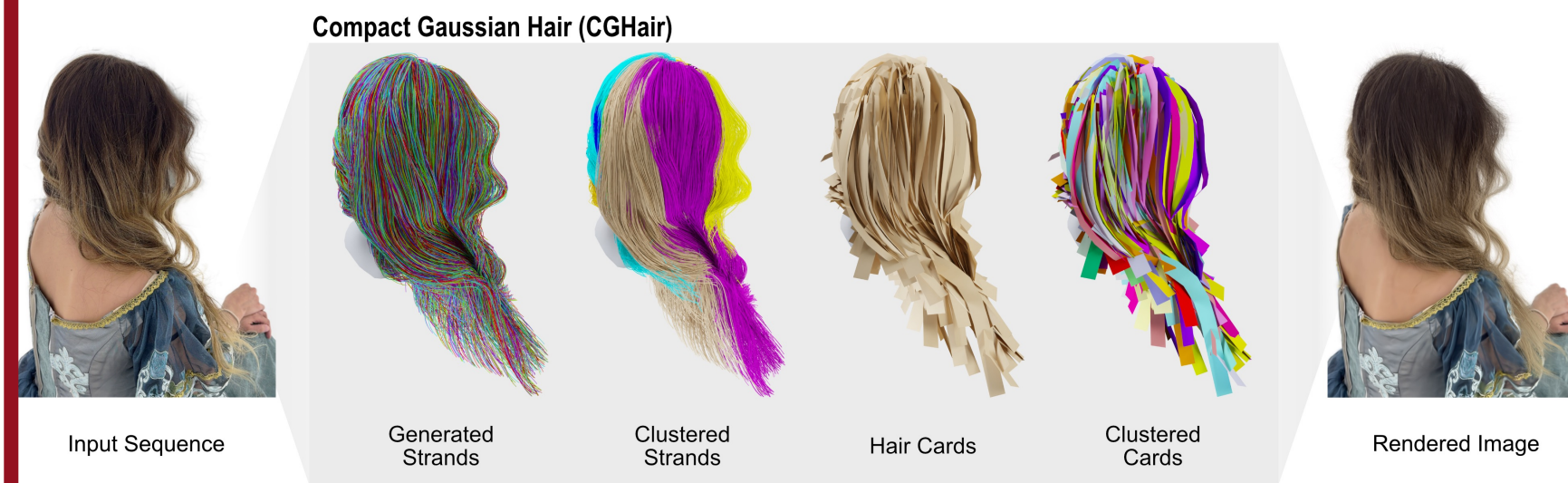
CGHair

Compact Gaussian Hair Reconstruction with Card Clustering

Haimin Luo^{1,2}, Srinjay Sarkar¹, Albert Mosella-Montoro¹, Francisco Vicente Carrasco¹, Fernando De la Torre¹
¹Carnegie Mellon University, ²ShangaiTech University

Goal

Reconstruct high-fidelity 3D human hair from multi-view video with a compact Gaussian representation — keeping rendering quality while slashing storage and reconstruction time.



Problem

- Redundancy: Existing 3DGS hair methods model every strand independently, ignoring the strong geometric and appearance similarity across a hairstyle.
- Reconstruction cost: Recovering strand-level geometry from multi-view video is slow, relying on weak structural priors and heavy diffusion or photometric regularization.

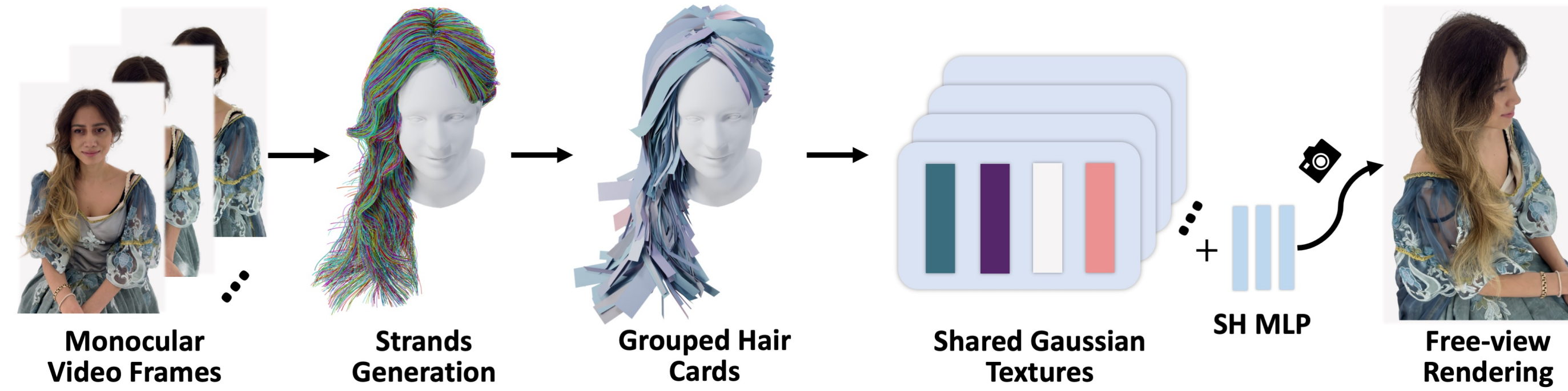
Highlights

- Hair-card-guided hierarchical clustering that removes strand-level geometric redundancy.
- A shared Gaussian texture codebook for hair cards, enabling scalable and consistent appearance modeling across structurally similar hair regions with **200x memory reduction**.
- An efficient method for reconstructing strand-level hair geometry from multi-view images, leveraging the PERM parametric model for a **4x speedup** over prior approaches.

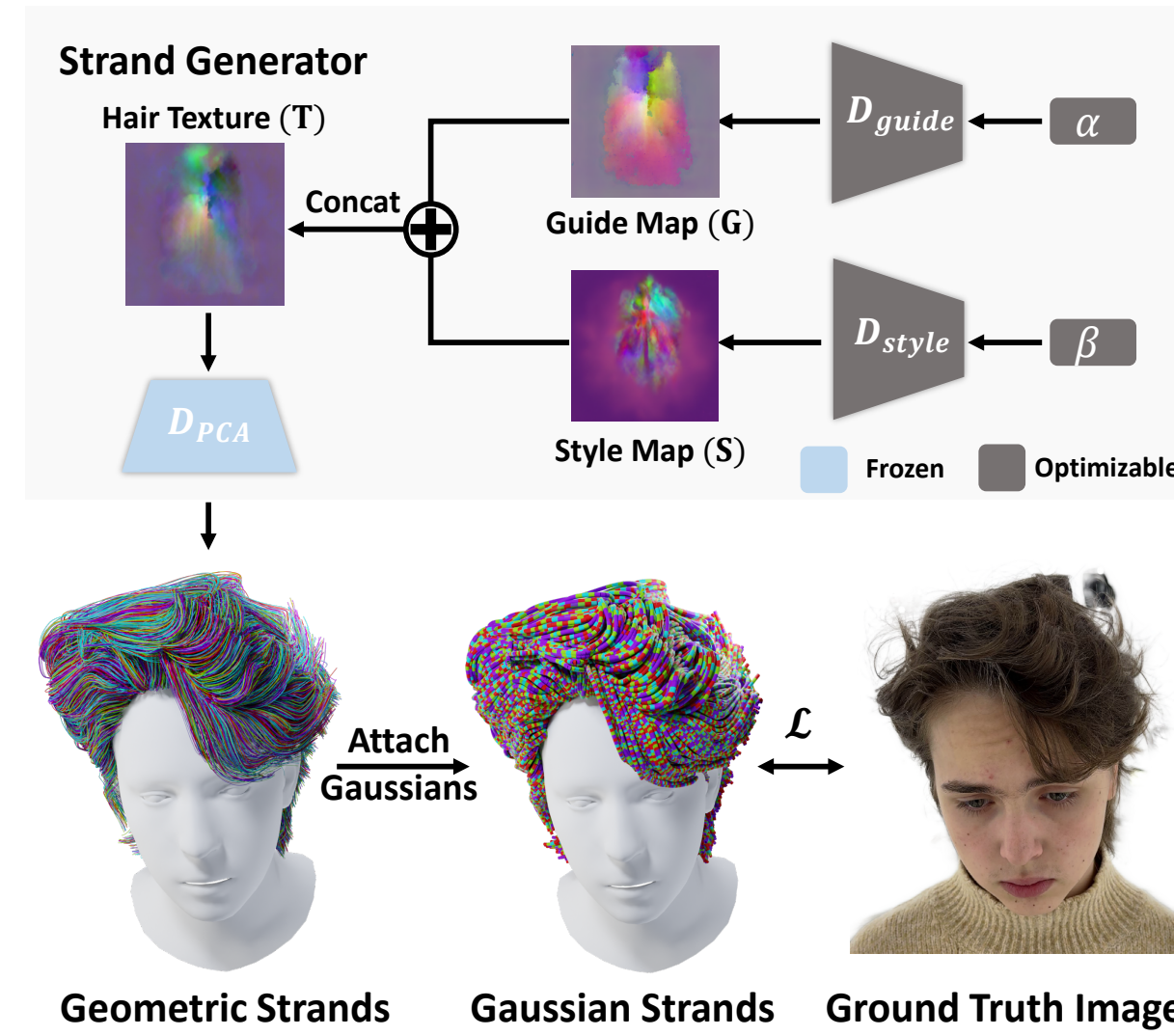
References
 [1] Gaussianhair: Hair modeling and rendering with light-aware gaussians, Haimin Luo, Min Ouyang, Zijun Zhao, Suyi Jiang, Longwen Zhang, Qixuan Zhang, Wei Yang, Lan Xu, and Jingyi Yu. arXiv preprint arXiv:2402.10483, 2024.
 [2] Monohair: High-fidelity hair modeling from a monocular video. Wu, Keyu, et al. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024.
 [3] - Human hair reconstruction with strand-aligned 3d gaussians. Zakharov, Egor, et al. European Conference on Computer Vision. Cham: Springer Nature Switzerland, 2024.
 [4] Perm: A Parametric Representation for Multi-Style 3D Hair Modeling. He, Chengan, et al. International Conference on Learning Representations 2025.

CGHair

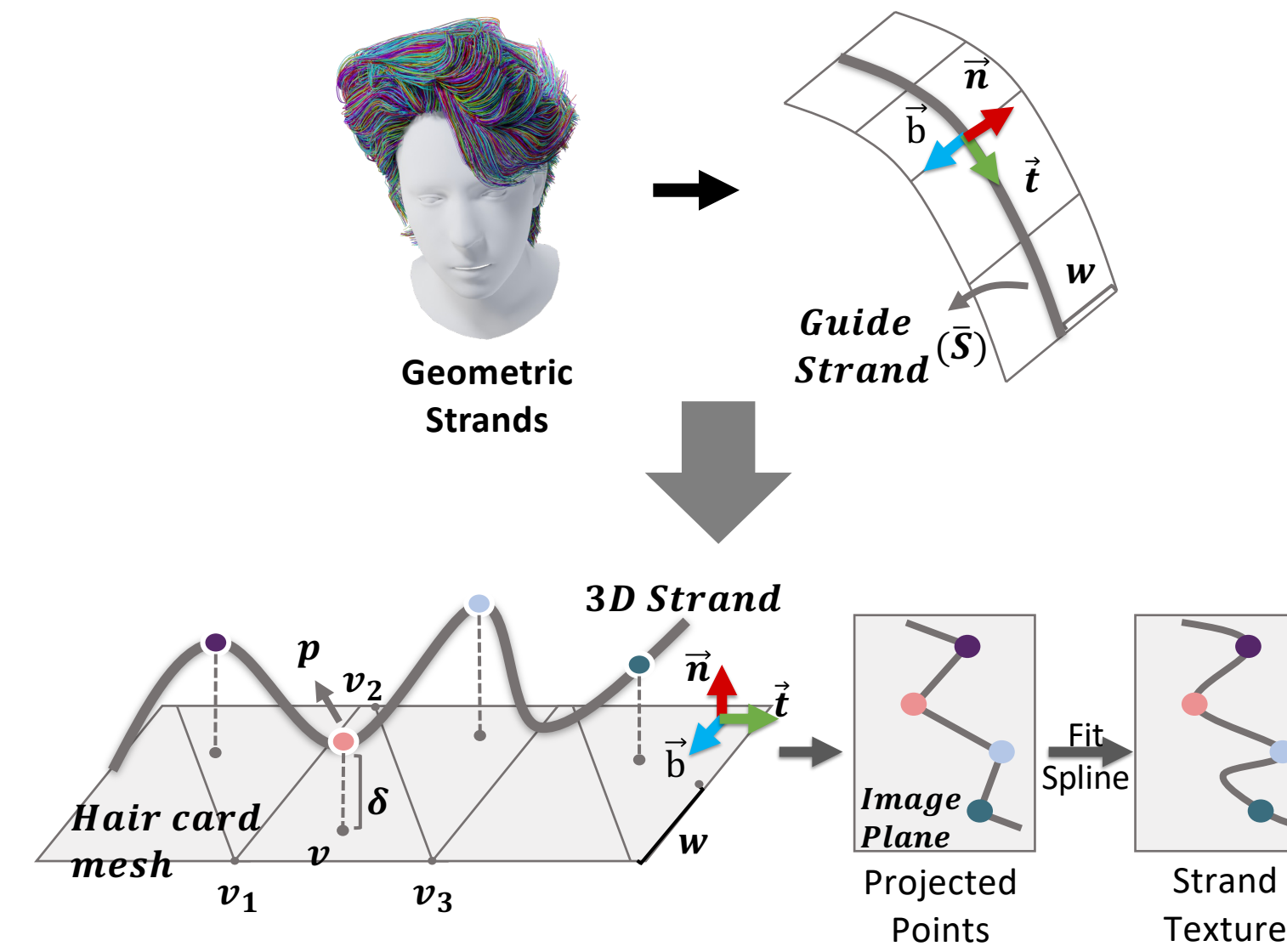
Pipeline



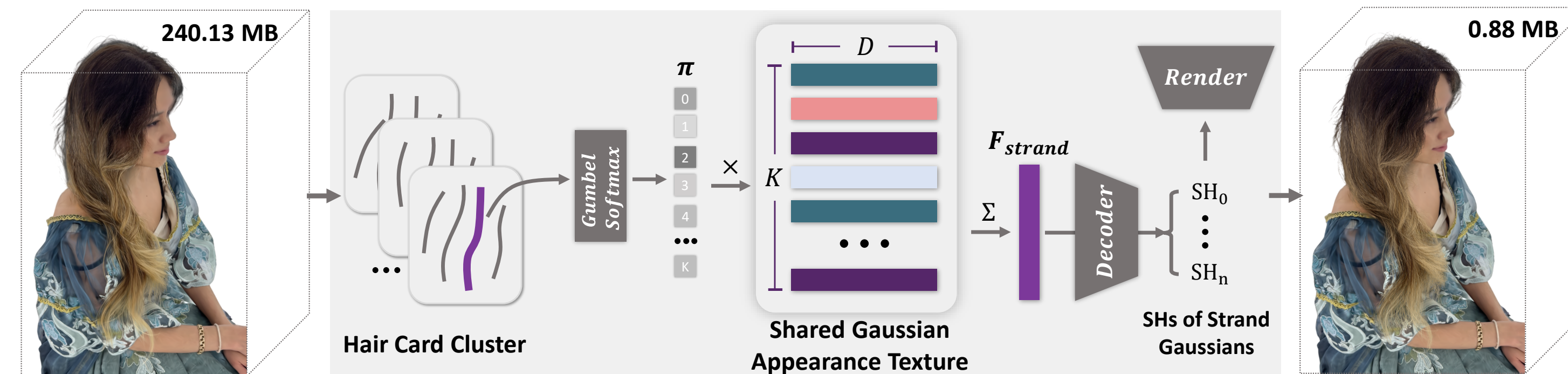
Strand Generator



Grouped Hair Cards

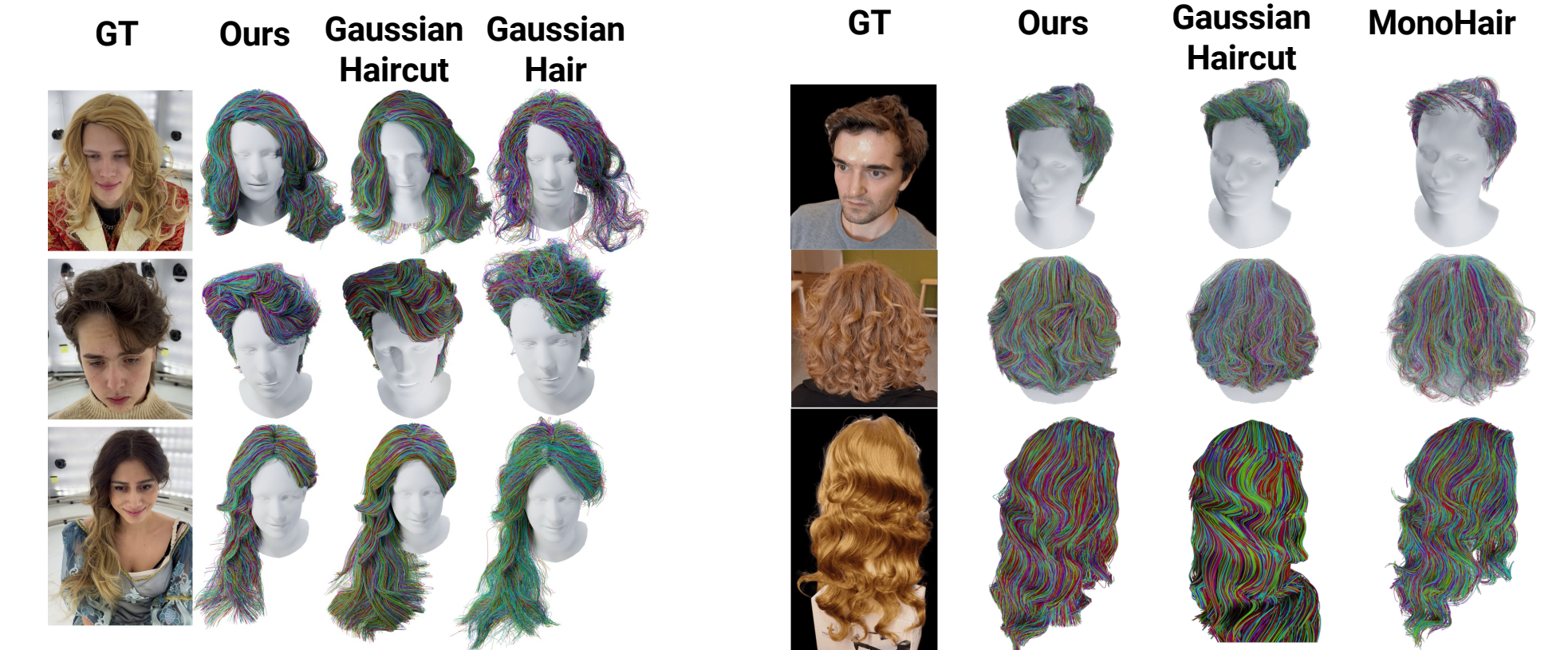


Shared Gaussian Appearance texture

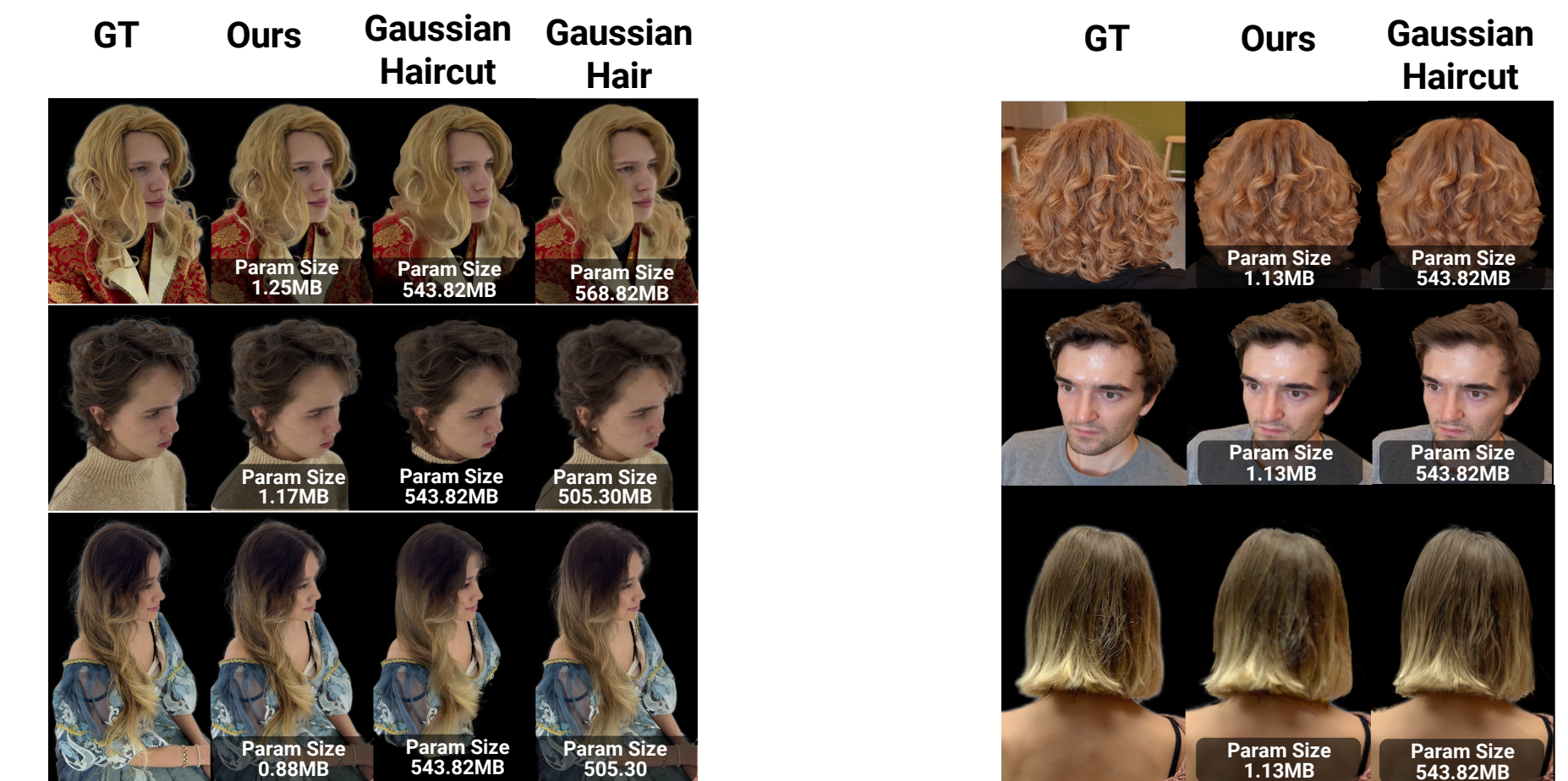


Results

Strands Generation



Gaussian Reconstruction



Animation

